

RHO5: note for expert review

Qianli Ma^{1,2}, Chao Wu¹ — 11 September 2026

¹Zhejiang University ²WUJIE AI

Email address (Qianli Ma): qianli.ma@zju.edu.cn

Purpose. We would appreciate a critical reading of the attached manuscript and supplementary index, particularly the global reduction and the interfaces between analytic lemmas and exact computational certificates.

Claim. For real 5×5 matrices, exact Gaussian elimination with complete pivoting has maximum element growth

$$\rho_5^{\mathbb{R}} = \alpha = 4.132517078632472854223346853277 \dots$$

The upper bound quantifies over every legal pivot path, including ties and singular termination. The constant is the unique root in $(4, 5)$ of the degree-61 polynomial identified by Chen, Edelman, and Urschel. Their attained lower bound is prior work; the manuscript supplies the global upper bound and hence the equality in their Conjecture 2.1 (arXiv version 2 numbering).

Proof route and reading order.

1. *Sections 3–5: analytic global entry.* After max-entry normalization, the first four pivot magnitudes are at most 4. Actual diagonal contractions preserve a nonzero fifth pivot and all earlier complete-pivoting conditions, producing one of two saturated tail geometries, D or X . Zero and tied boundary cases are treated explicitly.
2. *Sections 6–7: the two remaining domains.* The complete X bound is assembled from identified exact components. For each nonempty high-value fibre in the normalized range $p \geq 1$, the remaining negative D domain admits an attained maximum over six variables at a fixed seventeen-variable framework; no historical rank classification is an extra premise of this reduction.
3. *Sections 8–10: finite acceptance and assembly.* Exact certificates establish either infeasibility of $F \geq \gamma = 4132517/10^6 < \alpha$, or safety at α . Verified source bindings compose their domains into a complete cover. The accepted X and negative- D bounds then apply to every original matrix and combine with the attained lower bound.

Questions on which feedback would be especially useful.

- Do the analytic reductions retain the full quantifiers, including zero coordinates, ties, and common-matrix feasibility?
- Are the certificate rules and their stated source domains sufficiently explicit to support the global inference? What additional presentation or independent checking would you require?
- Is the claimed advance over the existing fifth-pivot result correctly positioned, and are important references or qualifications missing?

Evidence and present scope. The manuscript and supplementary index identify the exact polynomial, model, accepted components, and final coverage records. Historical exact acceptances are reused with explicit identities; the final composition is not a fresh replay of every earlier tree. The final B archives have been preserved locally and hash-checked; the full theorem also requires the identified upstream X payloads. Source, certificate downloads and step-by-step instructions are public at <https://github.com/hkjtsgmc79-boop/rho5-proof> (release v1.0.0). The guide states the remaining full-chain portability limitations. The material is supplied for expert feedback. A companion Lean development formalizes key analytic results and a conditional final theorem; full global safety still relies on the written arguments and exact certificates.

Related paper. Chen–Edelman–Urschel, *IMA Journal of Numerical Analysis* (2026), advance article drag039, published online 7 September 2026.

<https://doi.org/10.1093/imanum/drag039> Open manuscript: <https://arxiv.org/abs/2602.20390v2>.